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Convolutional Neural Networks accurately predict soil matric potential from soil, weather, and satellite-derived data

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ABSTRACT

Being able to predict soil moisture dynamics offers water managers the possibility to better plan irrigation events and prevent soil moisture deficits from reaching levels that reduce crop production. Machine learning (ML) model predictions can potentially assist farmers in managing irrigation water more efficiently. In this study, we aimed to assess the accuracy of a set of ML models in predicting soil matric potential seven days ahead in gravity-surface irrigated cotton paddocks and evaluate the models' performance for longer term predictions (14 days). The ML models used past soil moisture, weather, and satellite-derived crop-related data as features for the input parameters. Input data were structured in tuples that were organised following a 20-day 'window' approach that 'slid' one position forward after each training round. A convolutional neural network (CNN) model outperformed a Long Short-Term Memory, Dense Multilayer Perceptron, and Linear Regression model, the latter of which produced the least accurate predictions. The accuracy of the soil matric potential predictions with the CNN model was stable over time ($R^2 \geq 0.92$ and root mean square deviation ≤ 7.5 kPa). However, less accurate predictions were obtained for a short period after emergence and at crop senescence. This study demonstrates the feasibility of producing accurate predictions of soil matric potential in cotton fields at 0.20 m soil depth with a CNN model, which can be integrated into irrigation decision support systems.

1. Introduction

Soil moisture is an essential variable in soil hydrology. Soil moisture is of importance for agriculture and for understanding climatic effects on the global water cycle (Vereecken et al., 2022). In irrigated agriculture, soil moisture predictions are of great value for optimizing irrigation scheduling decisions in order to make more efficient use of the water resources (Adeyemi et al., 2018). This is mainly because having accurate information about soil moisture dynamics within coming days, offers water managers the possibility to better plan irrigation events in order to maintain soil moisture deficits at necessary levels to avoid crop stress.

Data-driven machine learning (ML) models have the capacity of learning relationships and patterns from data set inputs to accurately make predictions of parameters of interest (Alzubi et al., 2018). Such predictions can potentially assist farmers in managing crops and different aspects of the farm business more efficiently to increase profitability (Benos et al., 2021; Goel et al., 2022). Machine learning techniques have been investigated to estimate and predict soil moisture in agricultural land (Adeyemi et al., 2018; Carranza et al., 2021;

Chaudhary et al., 2022; Dubois et al., 2021; Greifeneder et al., 2021; Grillakis et al., 2021; Gumiere et al., 2020; Liu et al., 2021). Carranza et al. (2021) studied the capabilities of using a Random Forest (RF) ML algorithm for interpolation and extrapolation (prediction) of the root zone soil moisture over an agricultural catchment using in situ measurements from a network of soil moisture stations. Results showed the accuracy of soil moisture extrapolation with the RF model was low in comparison with the interpolation. Better short-term (one to four days) soil moisture predictions have been reported in other studies with RF, Support Vector Regression (SVR) and LSTM (Long Short-Term Memory) models using in situ soil moisture measurements as input parameters (Adeyemi et al., 2018; Dubois et al., 2021; Gumiere et al., 2020; Yu et al., 2021). However, the performance of these models significantly decreases over time (> four days) (Dubois et al., 2021; Gumiere et al., 2020). For irrigation scheduling purposes, water managers would benefit from accurate long-term (≥ 7 days) soil moisture predictions.

Machine learning has been used to directly estimate surface soil moisture (0.05 m depth) from satellite measurements (Abbes et al., 2019; Chaudhary et al., 2022; Greifeneder et al., 2021; Liu et al., 2021)

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as well as to upscale and downscale satellite remote-sensed surface soil moisture at regional, national, and global levels (Srivastava et al., 2013; Zhang et al., 2017). For irrigation scheduling, root zone soil moisture information is preferable to surface soil moisture because it better reflects plant water availability for transpiration. Brinkhoff et al. (2019) evaluated the performance of different models in predicting soil matric potential in a cotton crop at 0.20 m soil depth using weather forecast data, soil moisture and crop vigour information (normalized difference vegetation index, NDVI) obtained from satellite imagery. The most important soil moisture predictors identified by Brinkhoff et al. (2019) were the square of cumulative crop evapotranspiration (ET_c) minus rainfall and the square of growth degree days (GDD). In a recent study using data from the International Soil Moisture Network, Wang et al. (2023) showed that deep learning models (LSTM and convolutional neural network, CNN) outperformed ML models such as RF and SVR in predicting soil moisture. This was attributed to the higher capabilities of deep learning models in processing time series data. Recently, Maia et al. (2022) presented an approach where ML models were trained with in situ soil matric potential measurements and used to estimate soil matric potential between irrigation events in non-monitored areas of cotton fields from GDD, cumulative remotely sensed ET_c and cumulative rainfall data. This approach was based on the soil matric potential and ET_c relationship and it was proven useful to accurately estimate current soil matric potential in areas not monitored with in-field sensors. Models that could exploit such relationships described in Maia et al. (2022) between soil matric potential and ET_c for predicting soil moisture in coming weeks would be of great value for water managers (and for parameterising biophysical process agricultural crop models). This would allow optimizing irrigation scheduling decisions, and thus the water resources, in large farms with reduced investment in field monitoring equipment.

In this work, we aimed to assess the feasibility of accurately predicting soil matric potential in irrigated cotton fields from soil, weather and crop-related data using ML models. The specific objectives of the study were to assess the accuracy of the ML models in predicting soil matric potential a week ahead, evaluate the models' performance for longer term predictions (14 days), and identify the best performing model. Although used mostly for image processing (Patil and Rane, 2021), a CNN model was included in the study because it was hypothesised the convolution layer could effectively identify oscillations in the soil moisture time series signal caused by factors such as irrigation, precipitation, and weather fluctuations. We hypothesized that among the models assessed, the LSTM model would provide the most accurate predictions because of its reported suitability for learning from past observations with limited input data and its suitability for time series applications (ElSaadani et al., 2021; Wang et al., 2023).

2. Material and Methods

2.1. Characteristics of the cotton farms

The study was conducted using six complete cotton growing season datasets collected from three commercial cotton farms located in two cotton growing areas of Australia from 2019 to 2022. Table 1 summarises the most relevant characteristics (location, soil type, number of

bays monitored, bay size, and rainfall) of the monitored sites. More detailed information about the regions where sites were located can be found in Maia et al. (2022).

The irrigation system at all sites was a bank-less channel system where water is distributed across bays through furrows by gravity. Irrigation scheduling was made by water managers at each site based on their own experience except for the cotton farm located at Whitton during the 2020/21 and 2021/22 growing seasons when irrigation was scheduled based on soil matric potential thresholds. At this site (Whitton site), irrigation water must be ordered from a water delivery company at least two days in advance and thus, from flowering to maturity, when the crop is most sensitive to water stress, irrigation water was applied two days after soil matric potential at 0.20 m soil depth surpassed -40 kPa in any of the bays to avoid crop water stress.

2.2. Soil, weather, and plant measurements

Soil matric potential was measured in all monitored bays with Watermark sensors (Model 200SS, Irrrometer Company Inc., CA, United States; Fig. 1) installed in pairs at 0.20 m soil depth along with one temperature sensor (model DS18B20). Sensors were installed on hills, within the plant line soon after plant emergence.

Hourly data collected with the watermark sensors was temporarily stored by WiField loggers (Goanna Ag, Goondiwindi, QLD, Australia) and sent via Wi-Fi to the cloud. These data were then processed to obtain daily average soil matric potential for the entire cotton growing season.

Daily temperature data used to compute the GDD according to McMahon and Low (1972) with a base temperature of 12°C , rainfall, and reference evapotranspiration (ET_0) data were obtained from the nearest weather station to each site. The ET_0 was used to calculate daily ET_c of each monitored bay following the FAO56 approach (Allen et al., 1998). For this, the crop coefficient (K_c) was estimated across seasons from (10 m x 10 m) NDVI images of the bays obtained from the open-sourced Sentinel-2 satellite every week using the relationship between NDVI and K_c described in Trout and Johnson (2007). On days when no satellite images were available, NDVI was obtained by interpolation. Cumulative ET_c was calculated by adding daily ET_c for the period between irrigation events.

2.3. Predictive models

A dense Multilayer Perceptron (DMP), CNN and LSTM model (Fig. 2), which are described in the following subsections, were used along with the more popular multiple linear regression model to predict soil matric potential in cotton fields. All models were developed using the TensorFlow (v2.6.0) Keras API.

2.3.1. Dense multilayer perceptron

Dense multilayer perceptron is a Feedforward Neural Network (FNN) initially proposed by Hornik et al. (1989). The FNN algorithms use mathematical models called perceptrons (Rosenblatt, 1958) with the capacity to identify patterns in datasets from training. This property makes these algorithms of interest for regression applications (LeCun et al., 2015). The most common architecture of a DMP model comprises an input layer, where data are introduced to the algorithm, multiple

Table 1
Selected characteristics of the monitored sites located in NSW, Australia from 2019 to 2022.

Site location	Growing season	Monitored Bays	Bay size (ha)	Soil type	Cotton growing area	Rainfall* (mm)
Darlington Point	2019/20	3	5	Chromosol ^y	Murrumbidgee Valley	272
Whitton	2019/20	3	3–8	Chromosol ^y	Murrumbidgee Valley	272
	2020/21	3	3–8			340
	2021/22	3	3–8			509
Moree	2020/21	5	17–40	Vertosol ^y	Moree Plains Shire	645
	2021/22	6	14–42			562

* Rainfall recorded during the cotton growing season from October to May. ^y Isbell (2021).



Fig. 1. Illustration of the three data sources used for the soil matric potential predictions: (i) actual soil matric potential recorded with Watermark sensors at 0.20 m depth, (ii) on-site weather station, and (iii) site-specific satellite-derived crop evapotranspiration estimates from NDVI images. The NDVI map shown corresponds to the site in Moree in December 2021.

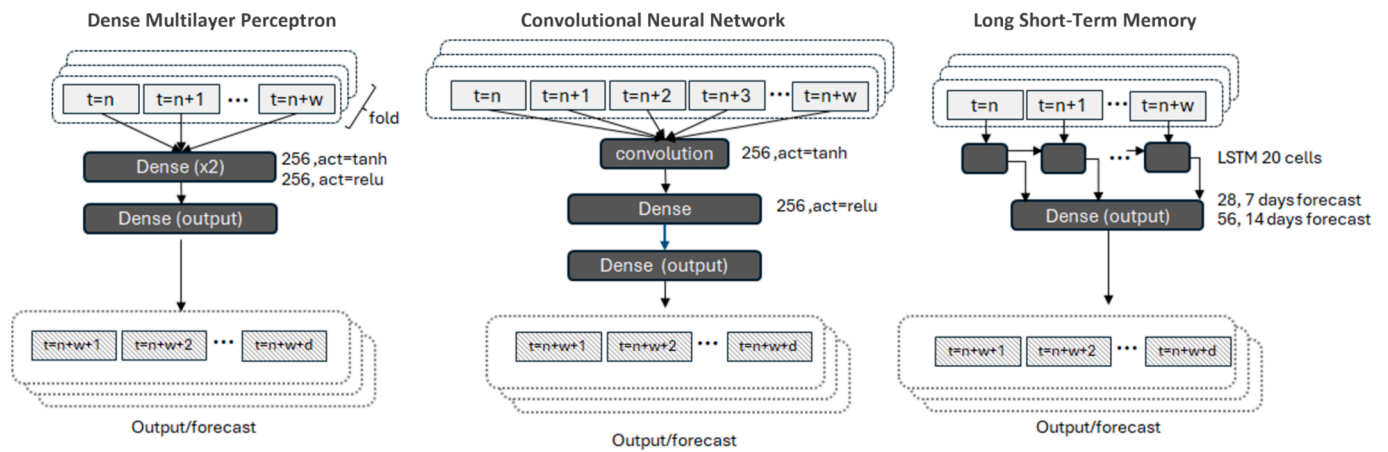


Fig. 2. Schematic diagram of the machine learning models. The same activation functions (act) and number of perceptrons in the dense layer were used in the Dense Multilayer Perceptron (DMP) and Convolutional Neural Network (CNN) models. The LSTM model had 20 memory gates. Training and testing processes were done using the same folds in all models. Each fold was composed of data from multiple ‘windows’, including data from all fields evaluated. *tanh* means hyperbolic tangent while *relu* means Rectified Linear Unit.

hidden layers where the patterns are processed, and an output layer where the results are presented (Irie and Miyake, 1988).

To predict the soil matric potential, the DMP model consisted of two hidden dense layers with 256 perceptrons in each of them (Fig. 2). Perceptrons of the first dense layer used the hyperbolic tangent (*tanh*) as the activation function while perceptrons in the second dense layer used the Rectified Linear Unit (ReLU).

2.3.2. Convolutional Neural Network

The CNN is another type of FNN algorithm. However, unlike the DMP, in the CNN algorithm the first layer contains convolutional filters that enable the extraction of features from the input data that can boost the pattern learning process. The CNN models use pooling layers which are used to reduce the dimensionality of the input data. Convolutional Neural Network algorithms are mainly used for processing matrix-structured data such as images. However, they have been reported to be effective in processing time-series data (Wang et al., 2023), which is crucial to predict soil moisture dynamics. In our study, the input parameters (described in section 2.4) were organized as a unidimensional matrix and the convolutions were applied in the same way using a kernel

to extract features from the dataset. The input parameters were normalized so all data had the same range. The activation function used in the convolution first dense layer was the hyperbolic tangent while the function used in the hidden dense layer was the ReLU with 256 perceptrons (Fig. 2).

2.3.3. Long Short-Term Memory

Long Short-Term Memory is a type of Recurrent Neural Network (RNN) (Song et al., 2024) initially proposed by Hihi and Bengio (1995) to solve the vanishing or exploding gradient problem commonly found in RNN models. The LSTM model uses feedback loops with a gating mechanism that enables reusing past data. This mechanism has been proven to significantly increase the ability to make predictions. Each layer of the LSTM model is composed of cells (like the perceptrons in the dense layer network). Each of these cells is composed of gates that are responsible for the memory of the model (Hochreiter and Schmidhuber, 1997). The configuration of the gates in each layer influences the memory and output of the model (Li et al., 2022), which in the case of this study is the predicted soil matric potential.

The LSTM model in the current study was composed of two layers.

The first layer had 20 feedback loops (Fig. 2). The second layer (dense output) was a dense layer with 28 cells for the seven-day predictions and 56 cells for the 14-day predictions (number of features * number of predicted days). The model was trained with the same activation and optimizer functions used in the ML algorithms previously described.

2.3.4. Training, and testing

Input data for the ML models was structured in tuples of four features $\langle \text{GDD}, \text{past soil matric potential}, \text{cumulative rainfall and cumulative } ET_c \text{ between irrigation events} \rangle$. Tuples were organized in a structure called a ‘window’ of a fixed size ‘ w ’ that slid one position forward after each training round (Fig. 3). An observation ‘window’ corresponded to collected data for the input parameters in the previous 20 days. The 20-day ‘window’ size was selected to ensure the observation included data between two irrigation events, and thus captured the (second degree) relationship between soil matric potential and cumulative crop evapotranspiration (Maia et al., 2022), aiming to optimise the models’ predictions. Reducing the ‘window’ size to less than 20 days could lead in some instances to an observation period not covering two consecutive irrigation events, thus not capturing the aforementioned relationship between soil moisture and cumulative evapotranspiration and the models providing less accurate predictions.

During the training process, algorithms were fed with the inputs and outputs (measured soil matric potential) for a ‘window’ w_i . The process was repeated for the ‘window’ w_{i+1} and so on until either the training algorithm reached 500 epochs or the root mean square error stabilised within 10 epochs. The Adam optimiser was utilised for training process in all models (Kingma and Ba, 2015). During the training process, the dataset composed of data from all monitored bays was shuffled to prevent the algorithms from learning the farmers irrigation practice behaviour at each site. That is, for example, irrigating cotton at regular intervals at specific stages of the crop. For validating and testing purposes, given an input ‘window’ w_i , the trained algorithm predicted the soil matric potential for the next K (seven or 14) expected GDDs (Fig. 3). Since input parameters were expressed in a daily basis, including GDDs, the predicted values then represented the soil matric potential predictions in the next seven or 14 days.

A different dataset was used in the testing process to assess whether the accuracy of the predictions was consistent with the models’ performance during the training and validation processes. Training data accounted for 70 % of the dataset while validation and testing accounted for 20 % and 10 %, respectively. All tests and models applied the K-Fold

cross-validation method (Bengio and Grandvalet, 2004). Each fold was composed of data from multiple ‘windows’, including data from all fields evaluated. The dataset was split into eight folds to have unbiased estimates and avoid overfitted models. The data set used in the study covered the whole cotton growing season of each year from late October until harvest (April/May), that is, from 0 to ~ 2000 GDDs. Only data identified as a sensor reading error (< 0.01 % of the whole data set) was excluded from the database.

2.3.5. Performance evaluation

The evaluation of the models’ performance in predicting soil matric potential was done considering data from all seasons and sites together. Two steps were taken to evaluate the performance of each algorithm, the validation and testing processes. The validation process consisted in comparing the coefficient of determination (R^2) resulting from the scatter plots of measured vs. predicted values, the root mean square deviation (RMSD), the Nash-Sutcliffe efficiency (NSE) and the mean absolute error (MAE) among models. The RMSD indicates the mean deviation of predicted values with respect to the measured values and was calculated according to Piñeiro et al. (2008). The NSE indicates how well the relationship between measured vs. predicted values fits the 1:1 line (the closer to 1 the better) and was calculated according to Moriasi et al. (2007). The MAE represents the average of the absolute errors.

3. Results

3.1. Seven-day predictions

The CNN was the model that best performed at predicting soil matric potential for seven days. The validation of the CNN model yielded an R^2 of 0.92 with 86.1 % of the predicted values within a difference of ± 10 kPa with respect to the measured soil matric potential (Fig. 4). Only 10.7 % of the predicted values were over 10 kPa difference with respect to the measured soil matric potential. The CNN had the lowest RMSD and MAE, and the highest NSE (Table 2).

The CNN model was followed in prediction performance by the LSTM model (Table 2). Most of the predicted values (78.7 %) with the LSTM model were within 10 kPa difference with respect to the measured soil matric potential (Fig. 4). However, a higher percentage (18.9 %) of the predicted values surpassed the 10 kPa difference in the LSTM model than in the CNN (Fig. 4). Both models CNN and LSTM tended to predict dryer soil matric potential than the measured (Fig. 4).

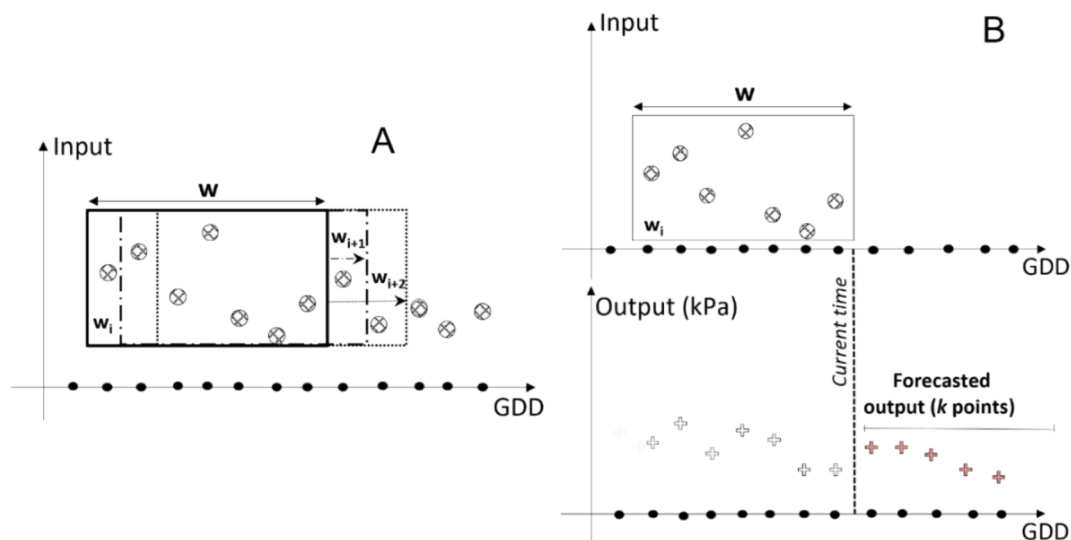


Fig. 3. Schematic of the sliding ‘window’ technique (A) and the validating and testing processes (B) with growth degree days (GDD) used as the temporal parameter. Points shown in each chart represent a 4-tuple.

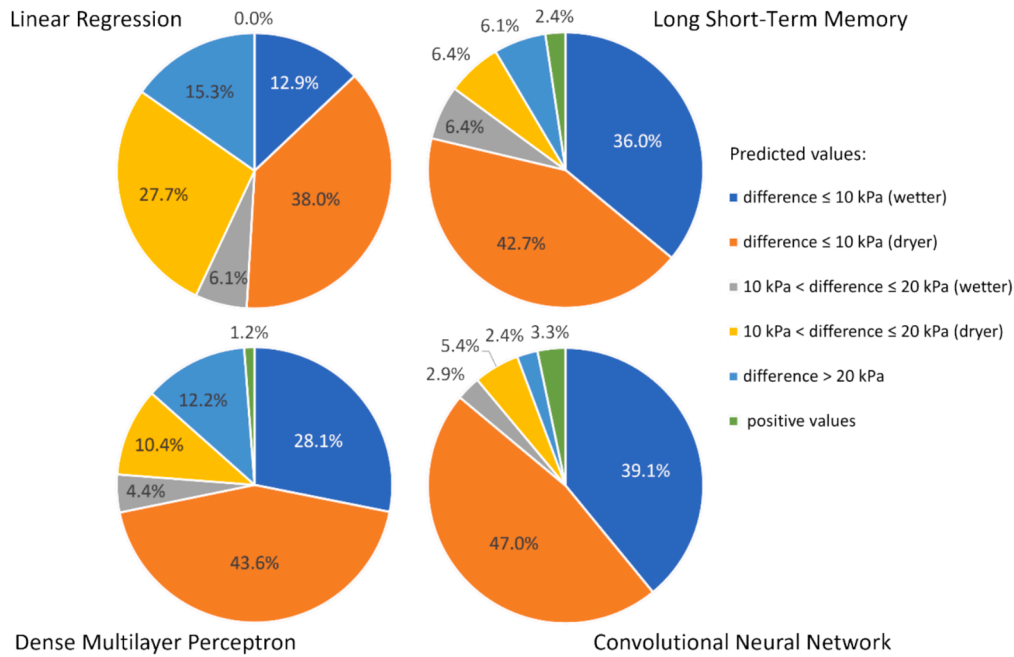


Fig. 4. Schematic distribution of the percentage of predicted values for seven days within 10 kPa, between 10 and 20 kPa and over 20 kPa difference with respect to the measured soil matric potential for each of the models assessed.

Table 2

Model performance indicators for the 7-day predictions. The R^2 represents the coefficient of determination for the relationship between predicted vs measured soil matric potential. The RMSD, NSE and MAE indicate the root mean square deviation, the Nash-Sutcliffe efficiency and mean absolute error, respectively.

Model	R^2	RMSD (kPa)	NSE	MAE (kPa)
Linear regression	0.39	19.60	0.38	13.15
Dense multilayer perceptron	0.65	14.87	0.64	9.03
Long Short-Term Memory	0.84	10.79	0.84	6.96
Convolutional Neural Network	0.92	7.53	0.92	4.87

Although with a large percentage (71.7 %) of predicted values within 10 kPa difference with respect to the measured values, the DMP model did not perform as well as the CNN and LSTM models (Table 2). The linear regression model performed the worst in predicting soil matric potential. The validation yielded a poor R^2 and almost half (49.1 %) of the predicted values surpassed the 10 kPa difference with respect to the measured soil matric potential (Fig. 4). The linear regression model had the highest RMSD and MAE, and the lowest NSE (Table 2). All the ML models except the linear regression predicted some positive soil matric potential values, although in a small percentage (≤ 3.3 %).

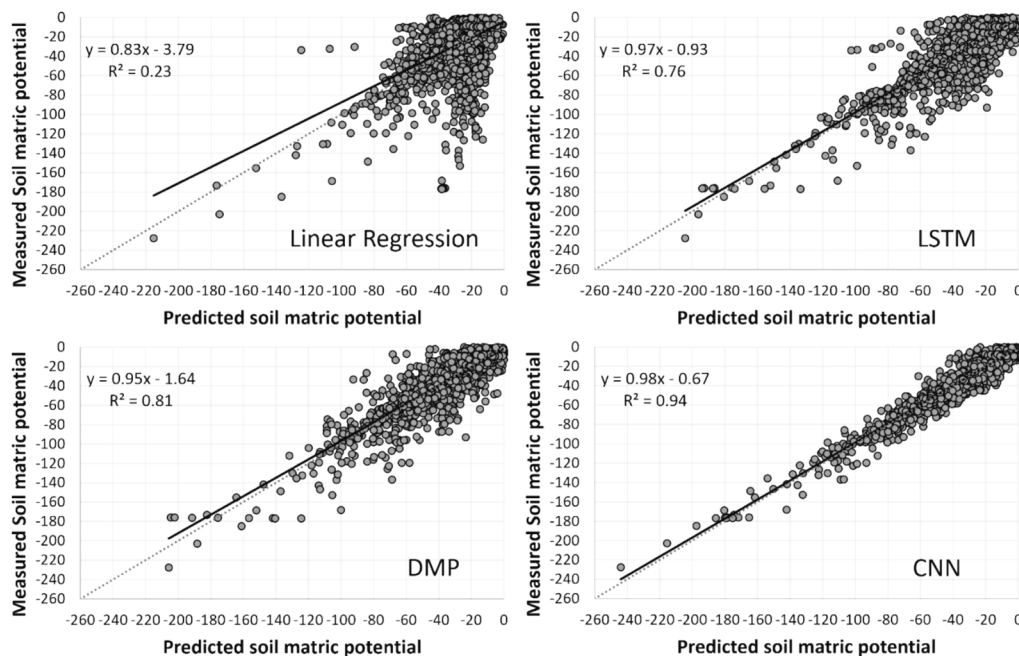


Fig. 5. Measured vs predicted soil matric potential for 14 days for each of the models assessed (n = 2800).

3.2. 14-day predictions

Similar results were obtained for the 14-day predictions when compared with the seven-day predictions in terms of prediction performance of the models. The most accurate 14-day predictions of soil matric potential were produced with the CNN model while the worst were produced with the linear regression model (Fig. 5; Table 3). More than 85 % of the predictions with the CNN model were within 10 kPa of the measured soil matric potential and only 1.6 % of the predictions were over 20 kPa difference with respect to the measured values (Fig. 6). The CNN model had the lowest RMSD with a similar value to that obtained for the seven-day predictions (Table 3). The CNN was also the model with the lowest MAE and highest NSE.

The measured against predicted soil matric potential relationship yielded a similar R^2 for the LSTM and DMP models although it was slightly higher in the latter (Fig. 5). Both models predicted a similar percentage of values within 10 kPa difference of the measured soil matric potential (66 % and 69 % with the LSTM and DMP models, respectively) (Fig. 6). However, the performance indicators were slightly better for the DMP model than for the LSTM (Table 2). As observed for the seven-day predictions, the CNN, LSTM and DMP models predicted positive values of soil matric potential on a few occasions with the DMP model predicting the largest percentage (Fig. 6). The linear regression model performed the worst at predicting soil matric potential (Table 3).

A comparison of the measured and predicted soil matric potential obtained with the CNN model for a GDD dataset composed of data from all the bays of a site and that covered the whole cotton growing season is provided in Fig. 7. The predictions followed closely the measured soil matric potential in a large part of the growing season from approximately 400 GDDs to 1800 GDDs. Less accurate predictions were obtained early (< 400 GDDs) and late (> 1800 GDDs) in the growing season. Plotting the soil matric potential and cumulative ET_c relationship for different irrigation events across the cotton growing season showed that a quadratic model provided the best fit to the curvature except at the last part of the growing season when a potential model fitted better (Fig. 8).

4. Discussion

4.1. Comparison among predictive models

The LSTM, DMP and CNN models outperformed the linear regression model in predicting soil matric potential. The better performance of the LSTM, DMP and CNN prediction models compared with the linear regression model could be related with two aspects. Firstly, the non-linear relationship that exists between soil matric potential and crop evapotranspiration (Fig. 8) that was recently reported by Maia et al. (2022). Secondly, the large dataset available in the current study for training the models (six-whole growing seasons' dataset), which is essential for the ML models to learn any existing relationship among the input parameters and the output (Alzubi et al., 2018).

In contrast to what was originally hypothesised, the CNN rather than the LSTM, was the model that performed the best at predicting soil matric potential (Table 2, Table 3). Good predictions were obtained with

the CNN model for the seven and 14-day period, indicating performance of the model was stable over time. The best performance of the CNN model in comparison with the rest of the models was most likely related to the convolutional layer of the CNN algorithm and its superior ability to learn soil moisture dynamics when there is significant variation in soil moisture (Wang et al., 2023). This most likely allowed for a better feature extraction of the time series data and more accurate predictions of the soil matric potential. The size of the input 'window' used in this study (20 days) ensured that at least a complete irrigation event was captured in the observation sample for the convolutional layer to detect the irrigation pattern. This was confirmed by the NSE index (> 0.92) obtained for both prediction periods assessed, which indicated the model was less influenced by noise than the other models.

The LSTM model followed the CNN model in prediction performance and outperformed the DMP and linear regression models when used to predict soil matric potential seven days ahead (Table 2). However, the DMP model performed slightly better than the LSTM in predicting soil matric potential 14 days ahead (Table 3; Fig. 5). The better long-term prediction capability of the DMP model compared with the LSTM could be explained by the unaltered memory structure (gates) of the latter regardless of the prediction period and its reduced number of layers (Fig. 2).

4.2. Performance of the CNN model across the growing season

Despite the good results obtained with the CNN model during most of the cotton growing season (400 – 1800 GDDs), less accurate predictions were achieved very early (0 – 400 GDDs) and late (> 1800 GDDs) in the growing season when cotton plants were either too small with a shallow root system or senescent and had a reduced or minimal daily crop water use (Fig. 7). Poorer soil moisture predictions (one day ahead) have been also reported for potato crops early and late in the growing season compared with in the middle (Dubois et al., 2021). Similar to what was reported in Dubois et al. (2021), this could be due to fewer samples available to train the model for this condition (very low or minimal plant water requirements) in comparison to the rest of the growing season when plants were actively up taking water from the monitored depth (0.20 m). The poor predictions obtained at the end of the growing season in our study could also be related to the change observed in the soil matric potential vs. crop evapotranspiration relationship at that stage of the crop in relation to the rest of the growing season (Fig. 8). In any case, soil moisture predictions at that time in the growing season when cotton plants are going to be defoliated are not relevant for irrigation scheduling decisions and very often are made based on the needs of the subsequent winter crop.

4.3. Comparison with similar studies

The prediction accuracy achieved with the CNN model for seven and 14 days was similar or even higher than the prediction accuracy reported in similar studies using ML models to predict soil matric potential one day ahead (Dubois et al., 2021; Gumiere et al., 2020; Wang et al., 2023). The observed stability in the prediction performance of the CNN model over time is an improvement with respect to what is reported in the literature (Dubois et al., 2021; Gumiere et al., 2020; Wang et al., 2023; Yu et al., 2021). In Wang et al. (2023), where a four-day sliding 'window' technique was used to organise data, the R^2 for the relationship between observed and predicted soil moisture at 0.20 m depth decreased from 0.94 to 0.74 when the CNN model was used to predict soil moisture one and seven days ahead, respectively. The better performance and stability of the CNN model for longer-term predictions in our study than in Wang et al. (2023), could be due to the larger observation 'window' (20 days) used in our study and to a better suitability of the input parameters (soil moisture vs crop evapotranspiration relationship) used in our study.

Table 3

Model performance indicators for the 14-day predictions. The RMSD, NSE and MAE indicate the root mean square deviation, the Nash-Sutcliffe efficiency and mean absolute error, respectively.

Model	RMSD	NSE	MAE
Linear regression	23.77	0.21	16.78
Dense multilayer perceptron	11.69	0.81	8.05
Long Short-Term Memory	13.05	0.76	9.16
Convolutional Neural Network	6.69	0.94	4.95

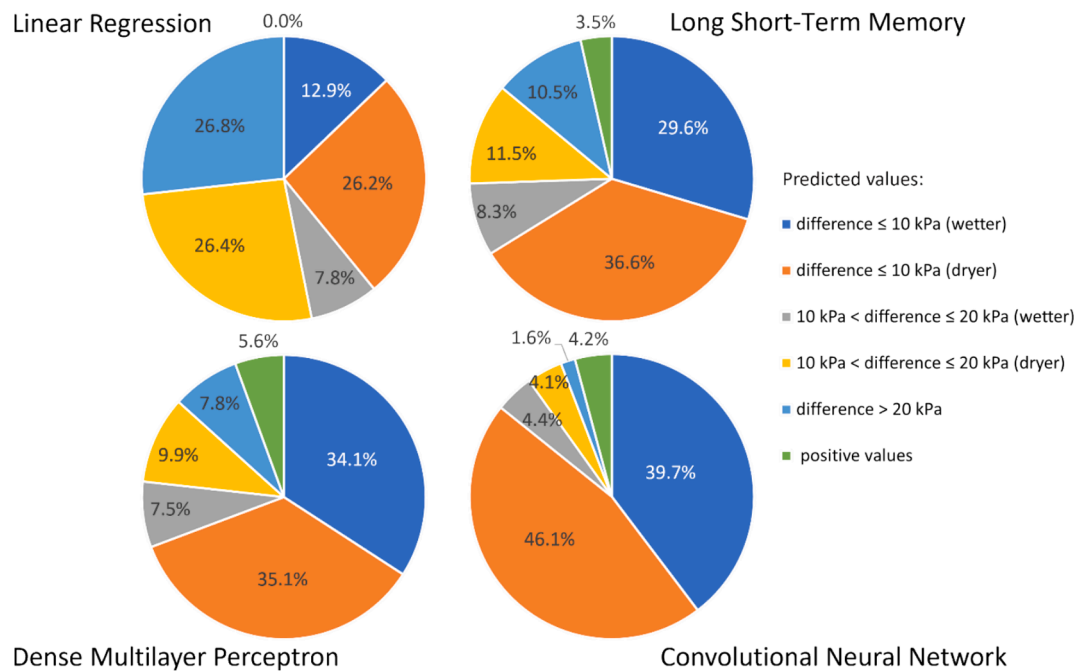


Fig. 6. Schematic distribution of the percentage of predicted values for 14 days produced by the models assessed within 10 kPa, between 10 and 20 kPa and over 20 kPa difference with respect to the measured soil matric potential.

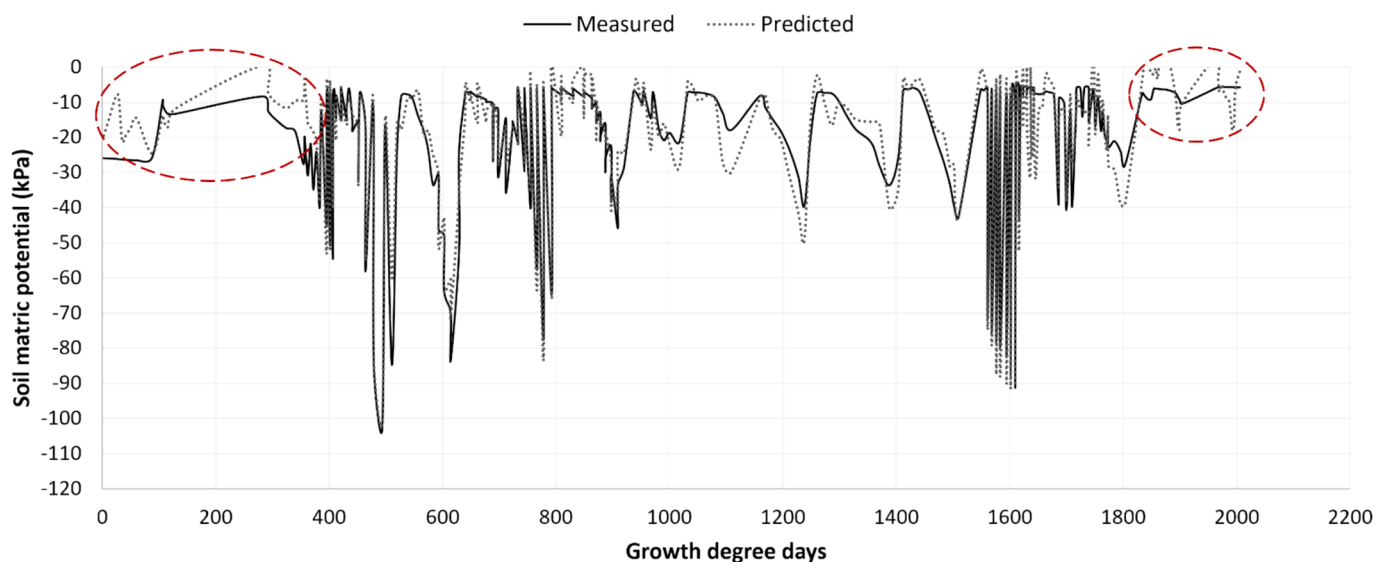


Fig. 7. Composition of measured soil matric potential from several bays of a site for the whole growth degree days dataset that covers a cotton growing season (from October to May) along with the predictions obtained with the CNN model. The areas delimited with a discontinuous line highlight the periods in the irrigation season when least accurate predictions were obtained with the CNN model.

4.4. Applicability of the predictive model

In cotton production systems under gravity surface irrigation, cotton producers can benefit from soil moisture predictions at bay level because irrigation events can potentially be automated to avoid detrimental soil moisture levels for crop production. Integration of the CNN model developed in this study into irrigation scheduling decision support tools such as IRRISENS (Filev Maia et al., 2020) that are used for irrigation water management at commercial scale would enhance the predictability of soil moisture and thus, the capacity of cotton producers to manage water more efficiently. This is particularly of interest in regions where irrigation water must be ordered to a water management company in advance (48 h in advance in one farm of this study) because

irrigation water can be ordered to be delivered at a desired soil matric potential.

5. Conclusion

This study demonstrates the feasibility of producing accurate predictions of soil matric potential for a week and 14 days using ML at a soil depth (0.20 m) where data can be effectively used to trigger irrigation events in cotton fields. This was achieved using past soil matric potential data, weather (cumulative rainfall and GDDs) and crop-related (satellite-derived crop ET_c) data as features for the input parameters. Longer prediction periods than 14 days were not considered because most of the irrigation events for the six entire-cotton-growing-season datasets

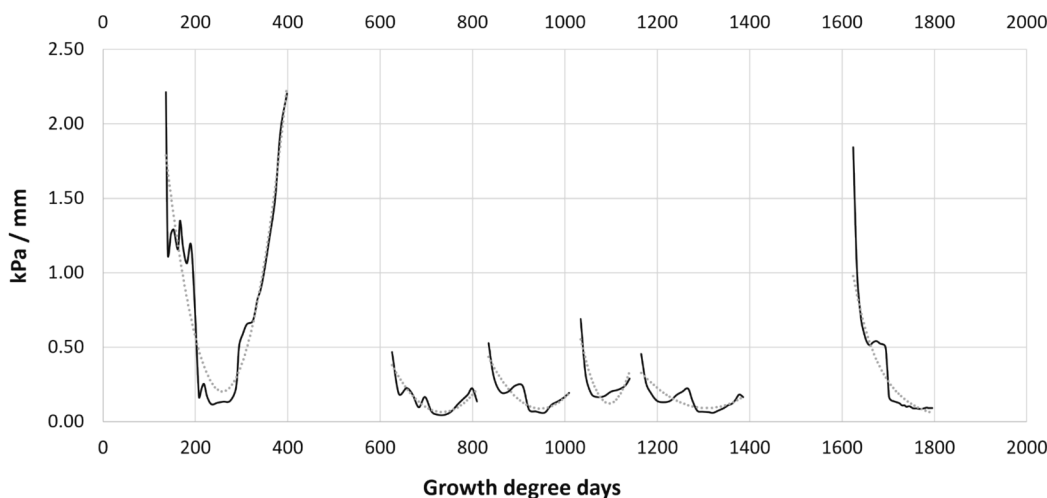


Fig. 8. Relationship between soil matric potential and cumulative crop evapotranspiration for six periods between irrigation events across the growth degree dataset that encompasses the entire cotton growing season. Up to 1400 growth degree days, the relationship between soil matric potential and cumulative crop evapotranspiration was better represented by a second degree function. In the last part of the growing season, a potential model fitted better.

occurred within eight and 16 days.

The CNN was identified as the ML model that most accurately (R^2 and $NSE \geq 0.92$, $RMSD \leq 7.5$ kPa) predicted soil matric potential. The convolution layer of the CNN model that acted as a filter for noise reduction enabled this model to predict soil matric potential more accurately than the DPM and LSTM models and maintain the prediction accuracy over time. More accurate predictions were achieved for the period between 400 and 1800 GDDs where cotton plants are actively up taking water from the soil than after emergence and at senescence when cotton water requirements are low and irrigation water scheduling decisions are of low importance.

Overall, the results from this work demonstrate there is potential for integrating ML models such as the one here proposed in irrigation decision support tools for optimising irrigation water and crop management in the cotton producing industry.

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CRediT authorship contribution statement

Carlos Ballester: Writing – original draft, Visualization, Methodology, Conceptualization. **John Hornbuckle:** Writing – review & editing, Methodology. **Brenno Tondato:** Writing – original draft, Software. **Rodrigo Filev-Maia:** Writing – review & editing, Software, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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